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1 Introduction

This section will provide a brief overview on smartphones, social media, and their eventual integration with one another. It will then provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are also mentioned. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development of the app.

1.1 Project Background

Social media, sometimes referred to as Social Networking Sites (SNS), have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with groups of individuals online from around the globe on a never-before-seen scale, allowing for friendships, communities and relationships to be built from virtually anywhere. Dating back to the days of MySpace in the mid-2000s, to social media networks such as X (formerly known as Twitter) and Facebook reaching out to billions around the globe, social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of social media presence (Shah 2021).

Similarly, smartphones have taken the world by storm, and have also become an important part of life for most people, even at early ages, with studies reporting that young people are receiving their first phone at the age of 10 (Chan, Yeo, and Handayani 2023). With the June 2007 release of Apple's "iPhone" and the release of the HTC Dream running the open source "Android" operating system in 2008, both Apple's iPhone and the myriad of devices that use Google's Android OS have cultivated a large shift in the world of online media (Goggin 2012), with it being reported in 2021 that 85% of adults in the United States own a smartphone (Ouellette and Rao 2022). Smartphones nowadays are used for communication via calling or texting through apps such as WhatsApp, for entertainment (gaming, TV, online content, etc..), for productivity, for learning, and a wide variety of other purposes.

It was only a matter of time before social media networks such as YouTube, X, and Facebook adapted their web interfaces to fit smartphones, and doing so has rewarded those networks with large gains in users for all, with it being revealed that 80% of content consumed on social media is through a smartphone (Islam et al. 2025). However, over the years, worries have been raised regarding a variety of issues that concern both smartphones and social media, particularly surrounding the

problem of individuals developing an overdependence on their smartphones and / or social media that negatively affects their day-to-day life. Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA), the issue has become a largely discussed and researched topic over the last 15 years with researchers looking into potential causes and correlations with other health effects.

New forms of media "harming the minds of the youth" isn't a new concept by any means. Going as far back to the invention of the radio, worries from parents about addictions related to radio, television, video games, the internet, and smartphones affecting young people have been prevalent, due to the youth's carelessness in regards to new technologies, but also due to the dystopian over-exaggeration on the dangerous effects that these at-the-time new technologies could have on society that these groups express (Abeele and Mohr 2021; Adorjan and Ricciardelli 2021).

While SA isn't classified the same way other addictions such as drug addiction are classified, the nature of it is instead classified as a behavioral addiction (King and Dong 2017), and a similar manner, SMA is considered as a subset of Internet Addiction (Hou et al. 2019), which itself is considered a subset of behavioural addictions (Chou, Condron, and Belland 2005), and while both are typically categorised as separate types of "addiction", it has been argued by some that SA may be correlated with SMA, due to smartphones themselves being enablers for addictive behaviors, including SMA (Kuss and Griffiths 2017).

The nature of SA and SMA has on young individuals have been extensively studied over the years, including why individuals may develop an addiction to both in the first place, and the effects that are correlated with both. For example, when looking at why some may be addicted to smartphones and social media, a study from Indian researchers Bhalerao et al. (2020) that analysed SA provided a multitude of possible causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology as a whole, therefore they may feel a preference to communicate over the internet as opposed to in-person, fueling the dwindling need for talking face-to-face with someone. When looking at the effects both are correlated with, in 2017, researchers in the United States King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to SA, in particular, they looked at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being. The results of their study found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, with social well-being as the only factor with no significant correlation. Following the results, they

concluded by mentioning that 3 out of their 5 hypotheses were consistent with past studies, while mentioning that regarding the hypothesis related to retention was "...a novel insight created from this study as current SA has not linked the consideration with college retention.".

Many researchers and developers have made attempts to study and provide treatment for the issue of SA and SMA, both as experimental studies on alternative treatments and as dedicated mobile applications. A 2024 India-based study (Seran 2024) aimed to look at using educational-based methods to reduce SMA and obesity. After answering a questionnaire, participants were split into a control group and a experimental group, the latter of which were then involved in educational activities such as lectures, brainstorming sessions, and exercises, among other activities. Regarding the results for SMA, participants were then scored on their SMA one, three, and six months after the initial study. The researchers found that there was a 21.2% difference in SMA scores between the control group and the group that partook in the educational-based intervention, with them concluding that it was an effective measure in reducing SMA. Other studies have looked to utilise different experimental methods, such as methods closely tied to psychology such as behavioural therapy. A China-based study from 2019 analysing SMA (Hou et al. 2019) developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, jotting down thoughts and emotions before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in their overall mental health, engagement in learning, sleep quality, and also saw a drop in social media usage.

For specifically mobile applications, in 2013, a group of German researchers Löchtefeld, Böhmer, and Ganev (2013), developed an android-based mobile application called "AppDetox" an app that's main objective was to provide a "medication" for SA, with the app allowing users to set restriction-based rules on apps that prevent the user from opening the app at specific times of the day, until the user removed the rule, and after a set amount of time. It also displayed information on their set rules, times they've broken said rules, and usage statistics on all their apps. Their analysis found that users mostly set restrictions on messaging apps such as WhatsApp and WeChat, and broke self-created rules at roughly a 7:1 ratio (7 times a rules was broken for every 1 rule set). Another app titled "AntiSocial" developed by Australian company BugBean for Android devices provided similar functionality to AppDetox, allowing users to restrict access to certain apps and provided app usage statistics but also added other functionality such as showing the user the app usage of other users that were within their age group, location, and occupation. It

also showed users how many times they picked up their device, and how long they used certain apps for (Michael 2017). According to the author of the journal entry, Katina Michael, the app ranked 3rd in downloads on the UK Google Play store mere days after launch, which "demonstrates the need for [AntiSocial.io]" (ibid.).

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user. Most literature that's within the topic of intervention apps only provide analysis on apps that deal with adjacent issues. For example, a 2021 study (Oliveira et al. 2021) reviewed the effectiveness of psychological intervention mobile apps to help with issues such as depression, anxiety, and a user's overall mental health, and while they concluded that the apps in the selected literature showed promise in reducing the aforementioned issues, none of the literature reviewed specifically looked at treating the issue of SA or SMA As well as this, most of these apps only provide the functionality to stop the user from using the app, but do not provide any sort of teachings for the user that can let them reflect on their usage. (ibid.).

1.2 Project Overview

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with smartphone and social media addiction by restricting usage of applications are in-demand, and a substantial amount of research and studies have been conducted aiming to suggest treatments to the issue by using experimental methods such as educational, and cognitive-behavioural therapy. However, very few apps have been researched, let alone developed and released with this approach in mind, with most consumer products on the app store opting to utilise app-restriction functionalities instead. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of their addiction and the potential causes of it, should they choose not to seek outside professional assistance, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology for example, young adults can be given opportunity to reflect on their addiction and learn how to treat the issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help treat smartphone and social media addiction

1.2.2 Project Objectives

The primary aim of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods. Two types of objectives have been identified, those being secondary objectives and primary objectives.

Secondary Objectives

Investigate existing mobile apps that use psychological methods to treat mental health issues Research into how existing mobile applications make use of psychological methods when treating issues such as depression and anxiety to understand how viable using mobile apps are to treat similar issues.

Investigate psychological methods used to treat smartphone and social media addiction
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate the contextualisation of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualised within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilised in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review.

Gather participants for evaluating the app and perform evaluation
Participants will be used when evaluating if the developed application reduces smartphone and social media usage more than existing apps that use restriction functionality. They will be given an overview of the app before being given a consent form to sign to consent to participating in the research. Participants will be split into

two groups, one will be using the developed app while the other will be using an existing app that has app restriction functionality.

Analyse the results and obtain a conclusion

Collect results from the evaluation and see if the app was more impactful at reducing smartphone usage than traditional apps. Then draw up a conclusion based on the results, and see if the problem statement was met.

1.3 Project Hypothesis

To test the hypothesis of this project, participants involved in testing the project will be split into two separate groups. One group will attempt to reduce their smartphone & social media usage by using a mobile application with usage tracking and psychological intervention methods, while the other will do the same, but by using a mobile application with traditional app restriction technologies instead. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within a mobile application will help to reduce the smartphone & social media usage in young adults more than existing apps which only use app restriction technologies.

2 Literature and Technology Review

2.1 Investigate apps that use psychological methods to treat things mental health issues

2.2 Investigate psychological methods used to treat smartphone and social media addiction

2.3 Investigate methods to track app usage

2.4 Investigate the contextualisation of SA and SMA in existing apps

3 Project's Primary Research Methods

This section will give a detailed overview on the methods that will be used to carry out the remainder of the project and provide a reasonable conclusion to the problem statement. This section will also outline what work has been completed up to this point and all remaining work that will be carried out over the project's lifetime.

3.1 Research Method and Justification

3.2 Develop the Mobile app (temp)

3.2.1 Development

3.2.2 Project Lifecycle

3.3 Usability Testing with users and Evaluation

3.3.1 Usability Testing

3.3.2 Evaluation

3.4 Ethical Issues

3.5 Preliminary Work

3.6 Remaining Work

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